Research Statement

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My research examines how machine learning and numerical methods can be combined for solid mechanics. My current work focuses on hybrid finite element–neural operator solvers based on non-overlapping domain decomposition. Earlier projects involved language model fine-tuning for mechanics, data-driven prediction of physical fields, and physics-informed neural networks. I am also interested in GPU-accelerated scientific computing and differentiable programming for structural optimization and inverse problems.

LLM Fine-Tuning for Mechanics

My first research project at Tongji University focused on instruction fine-tuning for question answering in mechanics of materials under Professor Ying Zhao's guidance. I constructed instruction datasets from textbooks and technical literature, covering stress analysis, constitutive relations, and failure theories. I organized the examples into instruction, input, and output fields and adapted the data loading code for training. Using low-rank adaptation (LoRA), I fine-tuned Qwen2.5-7B on these data. I publicly released two instruction datasets, the fine-tuned model, and its merged version on Hugging Face for other researchers to use.

Collaboration and Open Research

During the China International College Students' Innovation Competition, I worked with my teammates on local deployment of our fine-tuned language model. To accommodate limited GPU memory, we reduced the model's numerical precision and adjusted the checkpoint saving and loading procedures, enabling local inference on the available hardware. Our team received a national-level award in the competition. I have also documented my research projects in public repositories. I aim to make these materials useful to others by clearly describing the computational workflows and implementation choices.

ConvLSTM Prediction of Battery Material Fields

I subsequently studied conditional ConvLSTM models for data-driven prediction of lithium concentration and von Mises stress fields in battery materials, continuing under Professor Ying Zhao's guidance. I automated MATLAB–COMSOL simulations to construct a dataset in which each sample contained 25 time-aligned frames of each field, a grain-orientation map, and a C-rate map. I trained separate models in PyTorch for concentration and stress prediction, using a combined mean squared error (MSE) and structural similarity (SSIM) loss. At each prediction step, each model received five consecutive frames of its target field together with the two condition maps to predict the next frame. For multi-step autoregressive prediction, the predicted frame was appended to the input window and the oldest frame was removed, while the condition maps remained unchanged. Errors accumulated as model-generated frames replaced the reference frames in the input window, eventually producing visible distortion. To reduce the mismatch between training inputs and autoregressive inputs, I introduced scheduled sampling, gradually incorporating model-generated frames into the training windows. I compared the resulting concentration and stress sequences with COMSOL reference sequences to assess image errors and their accumulation over successive prediction steps. For the sequences

examined, the revised training produced lower image errors over the same prediction horizon and delayed visible distortion compared with training without scheduled sampling.

PINNs for Shock Capturing

My research with Professor Xianyang (Tom) Chen at Tongji University examined shock capturing for the one-dimensional inviscid Burgers equation using physics-informed neural networks (PINNs). I implemented a standard PINN in PyTorch, using automatic differentiation to compute the derivatives required for the partial differential equation (PDE) residual. The training objective combined PDE-residual, initial-condition, and boundary-condition losses, without using precomputed solution labels. To examine whether artificial viscosity could improve the representation of the sharp shock transition, I trained a second PINN with a fixed-coefficient artificial-viscosity term added to the residual. I evaluated both models against the analytical inviscid solution, comparing relative L^2 error, maximum pointwise error, shock location, and transition width. The artificial-viscosity model had a lower relative L^2 error but a higher maximum pointwise error than the standard PINN. I examined the predicted shock profiles alongside these metrics to distinguish overall agreement from local errors near the discontinuity. Because the added viscosity changed the governing equation, this comparison assessed agreement with the inviscid reference rather than the accuracy of two solvers for an identical PDE. The results therefore did not support judging the modification by a single error measure.

Hybrid Finite Element–Neural Operator Solvers

At Johns Hopkins University, I am developing a hybrid finite element–neural operator (FE–NO) solver for solid mechanics based on non-overlapping domain decomposition, under Professor Somdatta Goswami's supervision. The computational domain is divided into finite element and neural-operator subdomains connected through a shared interface. My collaborator and I initially explored predicting displacement fields and computing stresses from them, but this process amplified prediction errors. After discussing our tests, we chose to predict subdomain stresses directly. I sample interface displacement fields using Gaussian random fields and compute the corresponding subdomain stress fields with GPU-accelerated JAX-FEM simulations. These paired data are used to train Transolver in JAX to learn the mapping from interface displacement fields to subdomain stress fields. During coupling, the neural operator predicts stresses for the current interface displacements. The predicted stresses are used to compute interface reaction forces, which are passed to the finite element solver to update the displacements. The updated interface displacements then become inputs to the neural operator in the next iteration. I evaluate the coupled solver through stress-prediction accuracy, interface displacement compatibility, force balance, and convergence. When assessing computational efficiency, I distinguish the cost of a single neural-operator evaluation from the total cost of obtaining a sufficiently accurate coupled solution.

Stress-prediction errors were lower on held-out Gaussian random field (GRF) samples than on interface displacements encountered during FE–NO coupling. I compared the sampled displacement fields with those produced by the coupled solver and identified a distribution shift

between the training data and the actual coupling inputs. Using coupling-generated displacements as a reference, I adjusted the GRF length scale and amplitude to generate new interface displacement samples. I then computed the corresponding subdomain stress fields with JAX-FEM and retrained Transolver on the revised paired dataset. After retraining, the model maintained good stress-prediction accuracy on the GRF validation set and showed lower errors for coupling-generated inputs in cylinder and bracket tests. These results showed why validation on sampled data needed to be supplemented with tests inside the coupled solver. I also investigated convergence acceleration and interface relaxation strategies for the iterative coupling process. I am currently extending the framework to three-dimensional fiber-reinforced composites, using FEM to solve the surrounding matrix and sharing neural-operator weights across fiber subdomains. The matrix and fiber subdomains exchange interface displacements and reaction forces through the same iterative coupling framework.

Future Work

During my PhD, I hope to continue studying how machine learning and numerical methods can be combined for solid mechanics. My current work on finite element–neural operator solvers has deepened my interest in using learned models within numerical computations. I would like to learn about different coupling methods, explore which computations neural operators can perform, and understand how to assess the accuracy of the resulting solutions. I am also interested in model performance across mesh resolutions. I hope to explore whether an operator trained at one resolution can be used on finer or coarser meshes, and evaluate changes in prediction accuracy and computational cost. Working with JAX and JAX-FEM has drawn me to GPU-accelerated scientific computing and differentiable programming. I want to learn how numerical algorithms are implemented on GPUs and explore how neural operators can be combined with numerical solvers, using automatic differentiation to compute gradients for structural optimization and inverse problems. These directions require a stronger foundation in mechanics, numerical analysis, and optimization, which I hope to develop through doctoral training.

My long-term goal is to pursue research and development in scientific computing, including opportunities in industrial research teams at companies such as NVIDIA. I would like to continue exploring GPU-accelerated numerical methods and differentiable simulation, particularly their applications in structural optimization and inverse problems, and contribute to the development of scientific computing software. Doctoral training would help me build the theoretical foundation and research experience needed to develop and evaluate these methods.